

The Effect of Political Ideology on Violent Crime Rates

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Abstract

Does the political ideology of a state cause a change in violent crime rates? Using the dataset USArrests measures the murder, rape, assault, and urban population rates of states, we grouped the states into liberals and conservatives. On average, there was no significant evidence for a difference between the rates of murder, rape, and assault between liberal and conservative states. However, we found that we may have obtained the opposite results had our dataset been equipped with bigger sample sizes. We also found that on average, a higher urban population tends to increase the amount of rape within both liberal and conservative states, while there was no evidence to suggest that the same applied for murder and assault. We show these effects through regression models that map each violent crime onto the urban population of a state, and separate liberal and conservative states from each other. By determining causal effects, we are able to evaluate whether the results we attain are statistically significant or not. These findings improve the criminal justice literature, and can be used to determine whether certain conservative or liberal policies may cause more or less crime, or even suggest that there is no policy difference at all. Broadly, our study will help improve infrastructure and laws within states in order to lower crime rates.

Introduction

Research from the Manhattan Institute (Makridis VerBruggen) states that on county level, democrats tend to have more violent crime rates, while republicans are more violent on state levels, which shows that while there are varying results the data can be very easily manipulated in order to confirm a particular view. Crime rates generally increased during the COVID-19 pandemic, after the murder of George Floyd, which resulted in nationwide outrage. Conservatives quickly blamed the rising homicide rates on progressive policies such as defunding the police, while liberals pointed out that the lax gun policy may have caused the increase in homicides. Heritage Foundation (Anon), a primarily conservative organization, found that on average liberal states had higher homicide rates since 2002. CAP Action (Hall) compared large cities in blue vs. red states and confirmed that the cities were similar for numerous variables (but had difficulty controlling for racial demographics) and found that red cities had higher rates of gun homicides. The Center on Juvenile and Criminal Justice used CDC data and found that conservative states, and rural and small-town counties had been historically safer but had found higher homicide rates most years since 2018. As it stands, current studies within this literature are widely divided on whether the laws and policies within liberal and conservative states have a significant effect on their violent crime rates. This necessitates our study — by comparing the violent crime rates between liberal and conservative states, we use data analysis to come to a final conclusion on whether different policies significantly lower or worsen violent crimes, bringing an end to this debate.

Methods: Population Model

We employ one nationally representative study in our analysis, compiled within the USArrests dataset in RStudio. The dataset is composed of four main variables: rape, murder, assault, and urban population per 100,000 residents for each state in the USA. To answer our research question, we grouped the states into liberal and conservative based on their voting history in presidential elections. We acknowledge that there are confounding variables within our dataset that we can't control for, as there are non-political reasons that violent crime rates may go up such as poverty or psychological causes like malice and ill-intent. Therefore, we assume that there are no confounders to conduct our data analysis. To obtain our results, we estimate the population model:

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 S_i + \beta_3 X_i S_i$$

where Y_i is the outcome of the violent crime rate for an individual state i ; X_i is the measure of the urban population for a state i ; S_i is the indicator variable for whether a state is liberal or conservative, and holds values of 1 for liberal and 0 for conservative; $X_i S_i$ is the interaction term that shows the difference in violent crime rates between liberal and conservative states.

Methods: Regression Assumptions

In these models, we fit rape, murder, and assault onto the urban populations, specifying liberal states. To evaluate if there's a linear relationship, and assuming there are no confounders, a causal one, we created linearity, homoskedasticity, and normality plots to evaluate the regression assumptions.

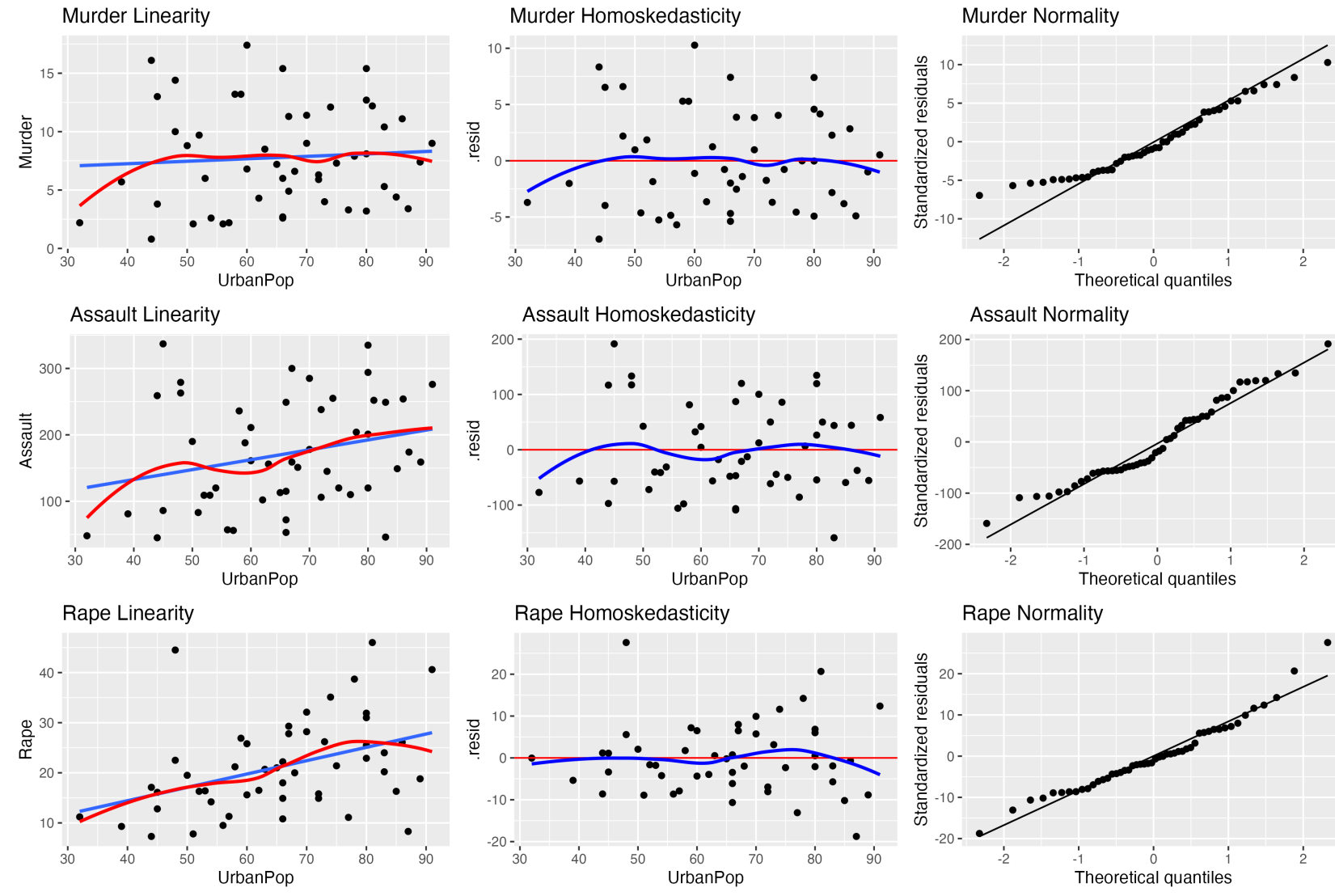


Figure 1. Regression Assumption Graphs

Regression Assumptions Analysis

In the first row, we have our regression graphs for the murder variable. While there is a slight non-linearity and heteroskedasticity on the ends of our graphs, the center is largely linear and homoskedastic, leaving the assumptions satisfied. The normality assumption is unsatisfied as there is a large deviation towards the bottom of the qq line. Independence is satisfied, as the murder rate of one state doesn't affect another's. Therefore, the assumptions are largely satisfied. For both the assault and rape variables, with the same previous patterns/explanations, our linearity, independence and homoskedasticity assumptions are satisfied. The qq plot assumption is also satisfied as there is little to no deviation. As the residual plots have no patterns as well, the assumptions are largely satisfied. As the residual plots have no patterns as well, the assumptions are largely satisfied. **As all the assumptions are largely satisfied, all the graphs show a linear and causal relationship under the assumption that there are no confounders.**

Results: Model Summary Tables

Below, we have displayed the summary statistics from our three population models:

	Murder	Assault	Rape
(Intercept)	7.327 (4.481)	102.564 (82.818)	4.840 (8.829)
UrbanPop	0.010 (0.073)	0.899 (1.342)	0.252 (0.143)
ideology	-2.815 (6.419)	-27.566 (118.656)	-2.804 (12.649)
UrbanPop × ideology	0.034 (0.096)	0.669 (1.778)	0.036 (0.190)
Num.Obs.	50	50	50
R2	0.012	0.078	0.170
R2 Adj.	-0.053	0.018	0.116
AIC	297.4	589.1	365.3
BIC	307.0	598.7	374.8
Log.Lik.	-143.714	-289.559	-177.628
F	0.185	1.296	3.149
RMSE	4.29	79.22	8.45

Results: Interpretations

When other variables are 0, the average rates for **intercepts** Murder, Assault, and Rape are 7.327, 102.564, and 4.840, respectively. For each 1 unit increase in the **urban population**, the average Murder, Assault, and Rape rates increase by 0.010, 0.899, and 0.252, respectively. When **ideology** is held constant at 1 for "Liberal", the average Murder, Assault, and Rape rates are -2.815, -27.566, and -2.804, respectively. For each 1 unit increase in **UrbanPop holding Ideology** constant at 1, the difference between liberal and conservative states for Murder, Assault, and Rape are 0.034, 0.669, and 0.036, respectively.

Murder and Assault: For the murder model, we found a p-value of 0.9061, which is higher than our alpha level of 0.05. Assuming that there was no difference, you'd obtain an observed value as or more extreme than 0.033610 90.61% of the time. For the assault model, we found a p-value of 0.2873, which is higher than our alpha level of 0.05. Assuming that there was no difference in the states, you'd obtain an observed value as or more extreme than 0.6687 28.73% of the time. Therefore, we fail to reject the null and there is no significant evidence to show that there is a difference between liberal and conservative states for murder and assault rates. **Rape:** We found a p-value of 0.03381, which is lower than our alpha level of 0.05. Assuming that there was no difference, you'd obtain an observed value as or more extreme than 0.03587 with a probability of 3.38% of the time. Therefore, we reject the null and there is significant evidence to show that there is a difference between liberal and conservative states for rape rates.

Outlier Evaluation

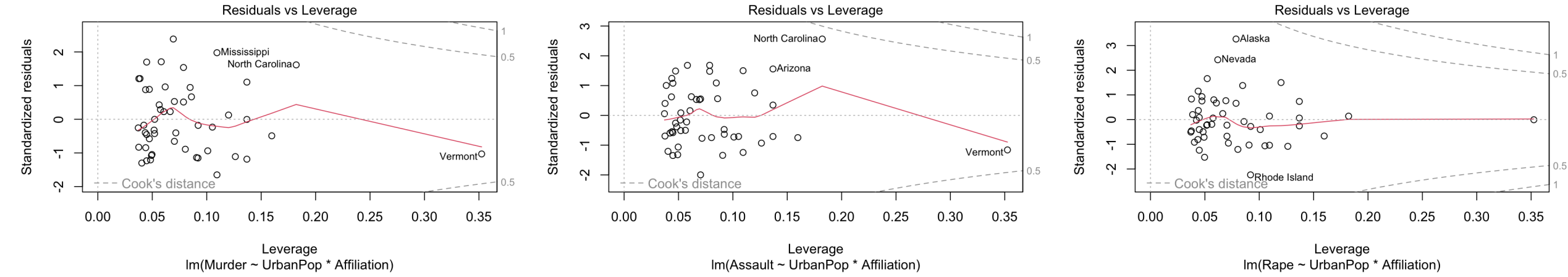


Figure 2. Outlier Maps

Our regression models showed low R-squared values, and the regression assumption graphs showed heteroskedasticity and deviation of the points along the ends of the graphs, which showed that the variance of errors was not constant. Due to this, we decided to analyze if this variance was caused by outliers. To do this, we generated leverage vs. residual plots for each population model. However, none of the data points exceeded the dotted 0.5 threshold on the graph, indicating that outliers were not a contributing factor to the low R-squared value. We concluded that the small sample size was likely responsible for the limited explanatory power of our model, rather than the presence of outliers.

Discussion: Significance of Results

Through our research, we found that the differences between liberal and conservative states for murder and assault were not significant, while the difference in the crime rate for rape was significant. There was a difference of 3.36% between liberal and conservative states for their murder rates, 66.87% for their assault rates, and 3.6% for their rape rates. Our confidence intervals for our first, second, and third population models are [-0.155, 0.222], [-2.815, 4.153], and [-0.336, 0.407] respectively.

Qualifications and Limitations: The study relies on standardized data sources (FBI, census, surveys) that might not capture all local variations or recent changes in urban demographics and political ideology. There also may be confounding variables such as socioeconomic status, or local law enforcement practices that could lead to biased results. A smaller sample size reduces the predictive value of our data and makes it harder to determine whether the results are statistically significant. **Causal Ordering:** This assumption states that changes in UrbanPop or Ideology cause changes in crime rates, and not the other way around. This is not plausible as the reverse can be true. **Consistency:** There should be a clear and consistent measurement of UrbanPop and Ideology in all states. This is not plausible as there can be variations in how these are defined or measured across different regions. **No Interference:** This assumption states that the crime rate of one state shouldn't affect the crime rate of another, which is plausible. **Unconfoundedness:** This assumption states that we have controlled for all important factors that could affect both the urban population (UrbanPop) and crime rates. If there are other factors that influence variables but are not controlled, it could lead to biased results. Therefore, to interpret our results causally, we assume there are no other confounding variables.

In our study's **internal validity** is very low because most of the causal assumptions are not satisfied. Underreporting of crimes such as rape or assault may also cause our study to have lower internal validity. In our study, the data on crime rates, urban population, and political ideology is sourced from official organizations such as the FBI and other governmental or academic sources. This data collection across various regions contributes to high **external validity**, meaning the findings are likely to be relevant and applicable to other contexts.

Substantive Significance: After increased homicide rates, conservatives and liberals questioned lax policies such as defunding the police or lack of gun control for causing this. However, our data shows that there is no significant difference between lib. and cons. states for both murder and assault. However, there is a significant difference between the states for rape, with liberal states having 3.59% more rape than conservative states, suggesting that there may be some policy problems causing this difference. **Statistical Significance:** We found that with a p-value of 0.033, there was statistical significance for the rape variable, however there was no statistical significance for both murder and assault. We believe that they may have been due to our smaller sample size within the dataset, there is a 66.87% difference between assault in liberal and conservative states; this suggests that a bigger sample size within our dataset may have yielded different results. Exploring this research question with bigger sample sizes within datasets for violent crimes may obtain far more statistically significant results that show differences between liberal and conservative states for certain crimes. Furthermore, expanding this study to control for more confounding variables such as racial demographics, political ideologies of local communities, etc, would help obtain far more accurate results as well.

References

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